

**FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION,
ISLAMABAD**

ANSWER KEY (Version 5)

Chemistry — Class IX — Annual Examination 2026

Syllabus: Ch 1 — Nature of Chemistry | Ch 2 — Matter | Ch 3 — Atomic Structure | Ch 4 — Periodic Table & Periodicity | Ch 5 — Chemical Bonding | Ch 6 — Stoichiometry | Ch 7 — Electrochemistry | Ch 8 — Energetics | Ch 9 — Chemical Equilibrium

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

Q1. Correct answer: **A) Environmental chemistry** 1 mark

Environmental chemistry studies chemical and toxic substances that pollute the environment. It is used to protect and purify water that has been contaminated by soil and dust, using methods such as **sedimentation, filtration, and disinfection**.

Q2. Correct answer: **D) Ca(OH)₂** 1 mark

From Table 2.2, calcium hydroxide's solubility **falls** from 0.173 g to 0.066 g per 100 g of water as temperature rises from 20°C to 100°C — the opposite of the usual trend shown by NaCl, KCl, and NH₄Cl (all of which increase). This unusual property is what makes Ca(OH)₂ useful as an antacid.

Q3. Correct answer: **B) Mass number** 1 mark

In an isotope symbol, the **mass number** (protons + neutrons) is written as a superscript to the upper-left of the symbol, while the **atomic number** (protons only) is written as a subscript to the lower-left.

Q4. Correct answer: **C) Irradiate and shrink cancer tumours** 1 mark

Radioactive **Cobalt-60** is commonly used to irradiate cancer cells in the hope of killing or shrinking tumours (Section 3.5.6, Uses of Isotopes). Iodine-131 is used for thyroid problems, Iodine-123 to image the brain, and Sodium-24 to trace blood flow.

Q5. Correct answer: **B) Br₂ + 2KI → 2KBr + I₂** 1 mark

Oxidizing power decreases down Group 17: **F₂ > Cl₂ > Br₂ > I₂**. Since bromine is a stronger oxidizing agent than iodine, Br₂ can displace I⁻ from KI. In the other options, a weaker halogen is being asked to displace a stronger one, which does not occur.

Q6. Correct answer: **C) He (1s²) is the exception; others follow ns²np⁶** 1 mark

Neon, argon, krypton, and xenon all have the general valence configuration **ns²np⁶**. Helium is the exception: its only shell is 1s² (just 2 electrons), satisfying the **duplet rule** rather than the octet rule.

Q7. Correct answer: **C) Ethanol** 1 mark

Ethanol is a **covalent** compound with a low melting point (−117°C) because only weak intermolecular forces need to be broken. NaCl, NaF, and MgCl₂ are ionic compounds held by strong electrostatic forces in a giant lattice, giving them much higher melting points (801°C, 996°C, 714°C).

Q8. Correct answer: **A) 60 g/mol** 1 mark

Molar mass of CO(NH₂)₂ = C(12) + O(16) + 2×N(14) + 4×H(1) = 12+16+28+4 = **60 g/mol**.

Q9. Correct answer: **C) 16** 1 mark

2(23) + 12 + 3X = 106 → 46+12+3X=106 → 3X=48 → **X = 16**. (This compound is actually Na₂CO₃, sodium carbonate, with X = oxygen.)

Q10. Correct answer: **B) Silver sulphide** 1 mark

Silver reacts with trace quantities of H₂S or SO₂ in the air or in food to form **silver sulphide (Ag₂S)**, which gives tarnished silver objects their blackish appearance.

Q11. Correct answer: **B) Chemical energetics** 1 mark

The branch of chemistry that studies energy changes (heat evolved or absorbed) in chemical reactions is called **chemical energetics**.

Q12. Correct answer: **C) An electric spark, no catalyst** 1 mark

Unlike the Haber process (iron catalyst) or the Contact process (vanadium pentoxide catalyst), nitrogen dioxide is produced from nitrogen and oxygen using an **electric spark**, with no catalyst required.

SECTION B – Short Questions (11 × 3 = 33 Marks)

2(i) Define medicinal chemistry and polymer chemistry... 3 marks

Medicinal chemistry studies the interaction between drugs and biological targets, and the development of new medicinal agents 1. **Polymer chemistry** studies polymers, their types, properties, and uses 1. Example applications: medicines are synthesized using knowledge of proteins/enzymes; nylon, polyethylene, and Teflon are common polymers used in everyday products 1. **OR** What is titration...: **Titration** is a technique used to determine the concentration of a solution, an application of **Analytical Chemistry** 1. Other applications of analytical chemistry: **forensic chemistry** (analysing body fluids, bones, fibers, and drugs) 1, and general use of instruments/methods to determine the composition and properties of substances 1.

2(ii) Differentiate concentrated vs dilute... 3 marks

A **dilute** solution contains a relatively small amount of dissolved solute, while a **concentrated** solution contains a relatively large amount of solute in the same amount of solvent 2. Brine is a concentrated solution because it holds a large amount of dissolved NaCl 1. **OR** Solubility curves KNO_3 vs NaCl...: KNO_3 's solubility rises steeply from about 20 g to nearly 150 g per 100 g water between 0°C and 100°C 1, while NaCl's solubility increases only slightly, from about 35 g to 39 g 1. KNO_3 is far more temperature-sensitive than NaCl 1.

2(iii) Define heavy water... 3 marks

Heavy water is water in which hydrogen-1 (protium) atoms are replaced by hydrogen-2 (deuterium) atoms 1. Its melting point (3.81°C) is higher than ordinary water's (0°C) 1, and its density (1.1044 g/cm^3) is higher than ordinary water's (0.997 g/cm^3) 1. **OR** Max electrons in sub-shells...: $s = 2$, $p = 6$, $d = 10$ electrons max 1.5. The M-shell ($n=3$) has 3s, 3p, 3d sub-shells, so its maximum capacity = $2+6+10 = 18$ electrons 1.5.

2(iv) Unknown element $3s^23p^4$... 3 marks

Valence electrons in $n=3$ shell → **Period 3** 1; $2+4=6$ valence electrons → **Group 16** 1; needs only 2 more electrons to complete its octet, so it is a **non-metal** (this is sulphur) 1. **OR** Unknown element $3s^1$...: $n=3$ → **Period 3** 1; 1 valence electron → **Group 1** 1; readily loses this electron, so it is a **metal** (this is sodium) 1.

2(v) Name four industrial catalysts... 3 marks

Any four: **Iron** – Haber process (ammonia synthesis) 0.75; **Platinum/palladium** – catalytic converters 0.75; **Nickel** – manufacture of margarine 0.75; **Platinum/vanadium pentoxide** – Contact process (sulphuric acid) 0.75. **OR** Metallic character trend...: Metallic character **increases down a group** (atomic size increases, outer electrons farther from nucleus, easier to lose) 1.5; **decreases across a period** (nuclear charge increases, shells stay same, valence electrons held more tightly) 1.5.

2(vi) NH₃-BF₃ coordinate bond... 3 marks

BF₃ has only 3 bonding pairs around boron and is **electron-deficient** 1. Ammonia's nitrogen has a lone pair which it donates entirely to boron 1, forming a **coordinate (dative) covalent bond**, represented H₃N→BF₃ 1. **OR** Covalent vs ionic melting points...: Covalent compounds (water, ethanol) are held by weak intermolecular forces, needing little energy to overcome 1.5. Ionic compounds (NaCl, MgCl₂) are held by strong electrostatic forces throughout a giant lattice, requiring far more energy to break - hence their much higher melting points 1.5.

2(vii) Industrial uses of graphite... 3 marks

Any three (1 mark each): **refractory bricks** (chemical inertness, heat stability); **electrodes/anodes** in electrolytic/metallurgical furnaces (conducts electricity); **moderator rods** in nuclear reactors (inertness); **carbon brushes** in electric motors; **bearings/piston rings/valves** (layered structure allows sliding). **OR** Silicon dioxide giant structure...: SiO₂ has a **giant covalent structure** where every atom is joined by strong covalent bonds throughout the crystal 1.5, so a huge number of bonds must be broken to melt it, giving very high hardness/melting point 1. Used in the production of **abrasives** 0.5.

2(viii) Moles in 31.6 g KMnO₄... 3 marks

Molar mass of KMnO₄ = 39+55+4(16) = **158 g/mol** 1. Moles = mass ÷ molar mass = 31.6÷158 = **0.2 mol** 2. **OR** Mass of 0.4 mol NH₄Cl...: Molar mass of NH₄Cl = 14+4(1)+35.5 = **53.5 g/mol** 1. Mass = moles × molar mass = 0.4×53.5 = **21.4 g** 2.

2(ix) Oxidation states Table 7.2... 3 marks

Group 1 = **+1**, Group 2 = **+2**, Group 13 = **+3** 1.5. Hydrogen is usually **+1**, but this is reversed to **-1** in metal hydrides (e.g. NaH) 1.5. **OR** Sn + 2AgNO₃ redox...: Sn: 0→+2 (loses electrons, oxidized) → **Sn is the reducing agent** 1.5. Ag: +1→0 (gains electron, reduced) → **AgNO₃ is the oxidizing agent** 1.5.

2(x) Key Points, chemical energetics... 3 marks

Any three (1 mark each): the study of energy changes in reactions is **chemical energetics**; a reaction releasing heat is **exothermic**, one absorbing heat is **endothermic**; ΔH is negative for exothermic and positive for endothermic reactions; bond breaking always **absorbs** energy, bond forming always **releases** energy; a catalyst lowers activation energy without changing the overall enthalpy of reaction. **OR** Chemical energetics def + 2 metals...: Chemical energetics is the study of energy changes during chemical reactions 1.5. Two metals mentioned in the introduction: **iron and copper** (also aluminium, steel) - produced using huge amounts of energy released by chemical reactions 1.5.

2(xi) Empirical/molecular formula 80%C/20%H... 3 marks

C: 80/12=6.67, H: 20/1=20; ratio 6.67:20 = **1:3** → empirical formula CH₃ (mass 15) 1.5. n = 30/15 = 2 → molecular formula **C₂H₆ (ethane)** 1.5. **OR** Balance C₂H₆+O₂...: **2C₂H₆ + 7O₂ → 4CO₂ + 6H₂O** 3 (check: C=4=4, H=12=12, O=14=14 - balanced).

SECTION C — Long Questions (4 × 5 = 20 Marks)

3(i) (a) Isotope reverse calculation **3** (b) Anion formation S, Cl **2**

(a) Total = $\text{RAM} \times 100 = 35.5 \times 100 = 3550$ **0.5**. Contribution of A = $75.77 \times 35 = 2651.95$ **1**. Contribution of B = $3550 - 2651.95 = 898.05$ **1**. Isotopic mass of B = $898.05 \div 24.23 = \approx 37.06 \text{ amu}$ **0.5**.

(b) S (Z=16) gains 2 electrons $\rightarrow \text{S}^{2-}$ (**config 2,8,8**) **1**. Cl (Z=17) gains 1 electron $\rightarrow \text{Cl}^-$ (**config 2,8,8**) **1**. **OR** (a) Three-isotope weighted average **3** (b) Protons/neutrons/electrons in electric field **2**: (a) $\text{RAM} = (24 \times 79 + 25 \times 10 + 26 \times 11) \div 100 = (1896 + 250 + 286) \div 100 = 24.32 \text{ amu}$. (b) Protons: +1 charge, ~1 amu, deflect toward negative plate **0.67**; Neutrons: no charge, ~1 amu, travel straight/undeflected **0.67**; Electrons: -1 charge, ~1/1836 amu, deflect toward positive plate **0.66**.

3(ii) (a) Empirical/molecular formula **3** (b) Balance $\text{P}_4 + \text{O}_2$ **2**

(a) C: $85.7/12 = 7.14$, H: $14.3/1 = 14.3$; ratio 1:2 $\rightarrow$ empirical CH_2 (mass 14) **1.5**. $n = 56/14 = 4 \rightarrow$ molecular C_4H_8 **1.5**.

(b) $\text{P}_4(\text{s}) + 5\text{O}_2(\text{g}) \rightarrow 2\text{P}_2\text{O}_5(\text{s})$ **2** (check: P=4=4, O=10=10 - balanced). **OR** (a) Ionic equation $\text{BaCl}_2 + \text{Na}_2\text{SO}_4$ **3** (b) Mass of MgSO_4 **2**: (a) Full ionic: $\text{Ba}^{2+} + 2\text{Cl}^- + 2\text{Na}^+ + \text{SO}_4^{2-} \rightarrow \text{BaSO}_4(\text{s}) + 2\text{Na}^+ + 2\text{Cl}^-$ **1.5**. Cancel spectator ions Na^+ and $\text{Cl}^- \rightarrow$ Net ionic: $\text{Ba}^{2+}(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) \rightarrow \text{BaSO}_4(\text{s})$ **1.5**. (b) Molar mass $\text{MgSO}_4 = 24 + 32 + 4(16) = 120 \text{ g/mol}$ **0.5**; mass = $0.3 \times 120 = 36 \text{ g}$ **1.5**.

3(iii) (a) Oxidation states Cr, N **3** (b) Redox agents $\text{Al} + \text{CuSO}_4$ **2**

(a) $\text{Cr}_2(\text{SO}_4)_3$: $3 \times \text{SO}_4^{2-} = -6$; $2\text{Cr} + (-6) = 0 \rightarrow \text{Cr} = +3$ **1.5**. NaNO_3 : $\text{Na} = +1$, $\text{O}_3 = -6$; $1 + \text{N} - 6 = 0 \rightarrow \text{N} = +5$ **1.5**.

(b) Al: $0 \rightarrow +3$ (oxidized) $\rightarrow$ **reducing agent = Al** **1**. Cu: $+2 \rightarrow 0$ (reduced) $\rightarrow$ **oxidizing agent = CuSO_4** **1**.

OR (a) Position of unknown element **3** (b) Transition vs s-block **2**: (a) Total electrons = $2 + 2 + 6 + 2 + 6 + 1 = 19 = Z$; valence $4s^1 \rightarrow$ **Period 4, Group 1, s-block, metal** (this is potassium). (b) (1) Transition elements show **variable oxidation states** (Fe: $+2/+3$) vs fixed oxidation state = group number for s-block (Na: always +1) **1**. (2) Transition elements form **coloured compounds** (Cu blue/green) vs colourless s-block compounds (NaCl) **1**.

3(iv) (a) Diamond vs graphite structure **3** (b) Uses **2**

(a) In diamond, each C bonds to 4 others in a rigid 3-D tetrahedral network with no free electrons $\rightarrow$ **extremely hard, electrical insulator** **1.5**. In graphite, each C bonds to only 3 others in 2-D hexagonal layers; the 4th delocalized electron moves freely within each layer (conducts electricity), and weak van der Waals forces between layers let them slide $\rightarrow$ **soft, slippery, conducting** **1.5**.

(b) Diamond: cutting/drilling tools (hardness); medical tools e.g. cataract surgery (hardness/precision) **1**. Graphite: electrodes in furnaces (conducts + heat-resistant); pencils/lubricant (soft, layered) **1**. **OR** (a) Buckminsterfullerene structure **3** (b) $\text{Ca}(\text{OH})_2$ antacid use **2**: (a) C_{60} consists of **60 carbon atoms** **1** arranged as **20 hexagons and 12 pentagons** forming a hollow, football-like sphere **1**; each carbon atom is bonded to **3 other carbon atoms** **1**. (b) $\text{Ca}(\text{OH})_2$'s solubility decreases from 0.173 g to 0.066 g per 100 g water as temperature rises from 20°C to 100°C **1**; this property is exploited to treat chemical burns and as an **antacid** to neutralize excess stomach acid **1**.

Chapters Covered: Ch 1 (Nature of Chemistry) | Ch 2 (Matter) | Ch 3 (Atomic Structure) | Ch 4 (Periodic Table) | Ch 5 (Chemical Bonding) | Ch 6 (Stoichiometry) | Ch 7 (Electrochemistry) | Ch 8 (Energetics) | Ch 9 (Chemical Equilibrium)

— End of Answer Key —